

Intermediate Scale Quantum (NISQ) setting [35] but not in the context of early fault-tolerant quantum computing.

In this paper, we introduce a framework that applies the quasiprobability method [31, 36] to explore the advantages of decomposing single-qubit  $Z$ -rotation gates into gates from the Clifford hierarchy [37]. A conceptual starting point is to consider the following two-step process: first, use magic state dilution [38] to convert a high magic resource into many low magic resources; and second, use quantum error mitigation to reduce any inherent noise or noise introduced in the dilution process. While each of these two steps could be cast as separate convex optimisation problems, it is more elegant and optimal to compress them. Therefore, rather than implementing these processes independently, our approach unifies them into a single optimisation problem, which we introduce as mitigated magic dilution (MMD). Specifically, we use convex optimisation to find the optimal sample complexity of performing small-angle single-qubit rotations from noisy encoded magic states.

There has been significant progress in the preparation of encoded magic states. For example, in Ref. [39], it was shown that encoded magic states can be prepared with a logical error rate of  $\sim 0.4 \times 10^{-3}$  under the assumption of ideal single-qubit operations and depolarised two-qubit gates with 0.1% error rate using post-selection. Moreover, there are several improvements to the state preparation of non-stabiliser states for arbitrary small-angle rotations that could offer further advantages to the results presented in this paper, which we examine in Section V.

We compare our framework against a baseline classical approach where small-angle single-qubit rotations are decomposed into Clifford operations, similar to Refs. [30–34]. Notably, we demonstrate a polynomial advantage, the magnitude of which depends on the quality of initial magic state preparation. We quantify this advantage in terms of the polynomial degree of magic resource saving of our method. For encoded magic states prepared with 1% dephasing noise, this saving is better than cubic, while for 0.1% dephasing noise we find a saving of degree approximately 11.43. Moreover, we show that our method can improve upon a state-of-the-art classical simulator, the sum-over-Cliffords stabilizer extent method [34], offering a 2.37 degree of magic resource saving.

To evaluate the practical benefits of this approach, we study the resource requirements to simulate the time evolution of the 2D Fermi-Hubbard model. The 2D Fermi-Hubbard model [40, 41] is of notable interest in the early fault-tolerant regime [6, 42] due to its importance in condensed matter physics (e.g., to understand high-temperature superconductivity [43] and the Mott metal-insulator transition [44]), as well as its simplicity arising from its highly regular lattice structure. Our analysis in Section IV demonstrates that MMD is a more resource-efficient method than direct gate synthesis, requiring fewer magic states in total. Moreover, the expected number of magic states per sample is significantly

smaller, and therefore MMD is particularly amenable to early fault-tolerant devices.

Even when quantum computers have an asymptotic speedup over classical computing they can fail to have an in-practice speed-up for example problems of a relevant size [45], and so it is crucial to make such comparison. Here, we consider the example problem of a Fermi-Hubbard model with a  $6 \times 6$  square lattice and evolution time  $t = 0.25$  as a strong candidate for quantum advantage. We find that our MMD method requires a circuit with only 1037 non-Clifford gate teleportations (on average) and  $5.34 \times 10^6$  samples. Even with error correction overheads, a sample per second is a conservative runtime estimate, taking a single quantum computer 62 days for all samples. For the same calculation, the sum-over-Cliffords stabilizer extent method would require a  $5.18 \times 10^{17}$  seconds runtime, which would take  $1.64 \times 10^4$  years to complete when assuming a million classical processors in parallel. Thereby, our MMD protocol facilitates a significant reduction in runtime over the the sum-over-Cliffords classical simulator for conservative time evolution and lattice sizes of interest. Combined with a Pauli-based model of computation [46], such an application would require logical error rates of approximately 1 part in a million, the so-called MegaQuOp regime [9].

This paper is structured as follows. In Section II we briefly summarise the quasiprobability method. In Section III, we introduce our framework in two steps, working in the channel representation of all unitary gates  $U$  throughout, which are denoted as  $\mathcal{U}(\cdot) = U(\cdot)U^\dagger$ . We first present the application of the quasiprobability method to target  $Z$  rotation channels over diagonal Clifford +  $\mathcal{T}$  channels in Section III A using a linear combination of channels decomposition. We then generalise this to decompositions over diagonal Clifford +  $\mathcal{T}^{\frac{1}{n}}$  channels in Section III B. Finally, in Section IV, we demonstrate the practical applicability of our framework for a second-order Trotter simulation of the Fermi-Hubbard model with comparison to gate synthesis.

## II. QUASIPROBABILITY METHOD

The motivation behind the quasiprobability method is that an ideal quantum operation can be decomposed into a basis set of noisy operations. Let  $\mathcal{U}^t(\rho) = U^t \rho U^{t\dagger}$  be the ideal target unitary channel of a quantum operation  $U^t$  and let  $\{\mathcal{U}_i^n\}$  be a set of channels corresponding to the noisy operations that can be performed on a given quantum hardware. The noise on these operations would typically be characterised using a tomography procedure [26, 47]. We can then write the following decomposition, hereafter referred to as the linear combinations of channels (LCC) decomposition

$$\mathcal{U}^t = \sum_i x_i \mathcal{U}_i^n \quad (1)$$